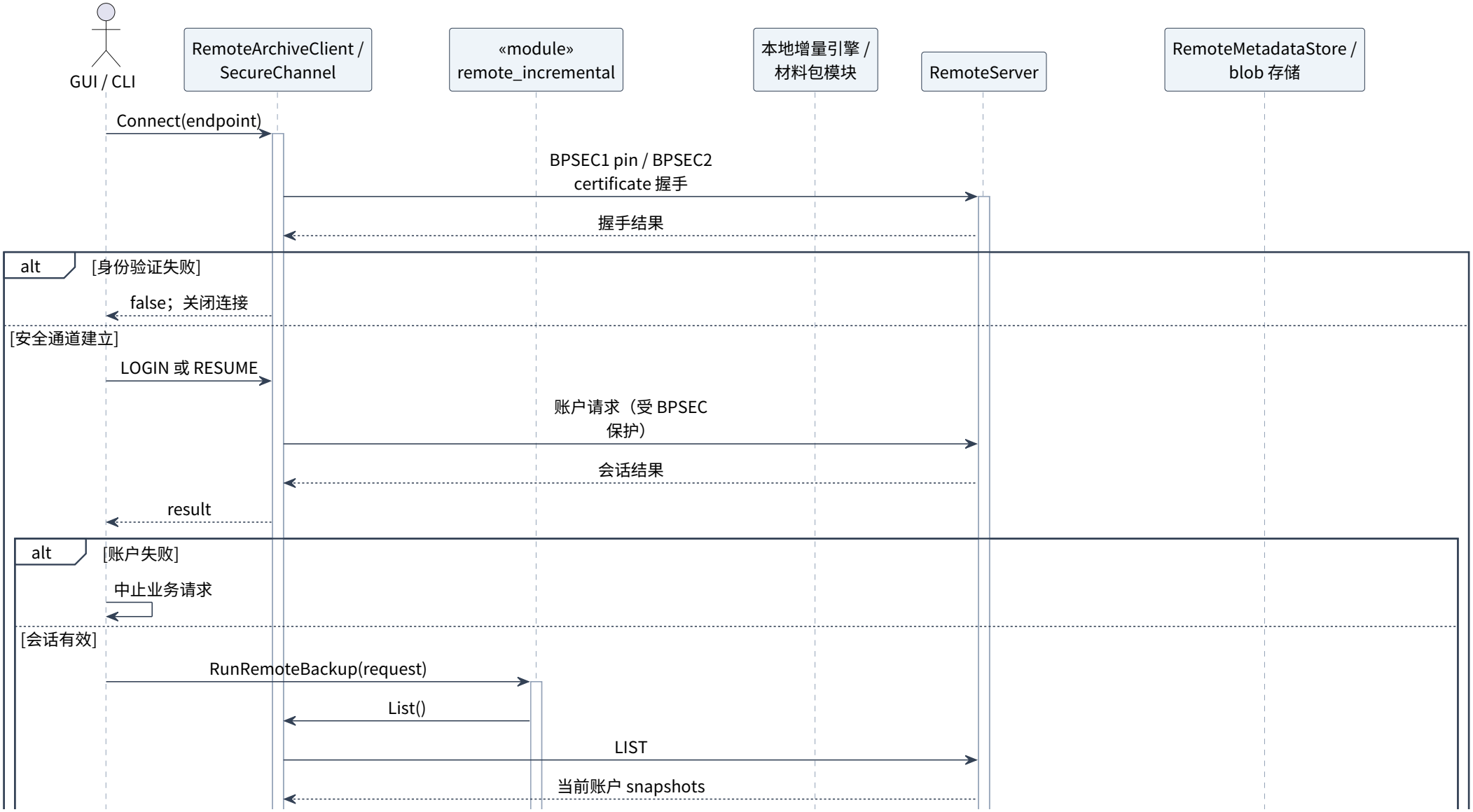
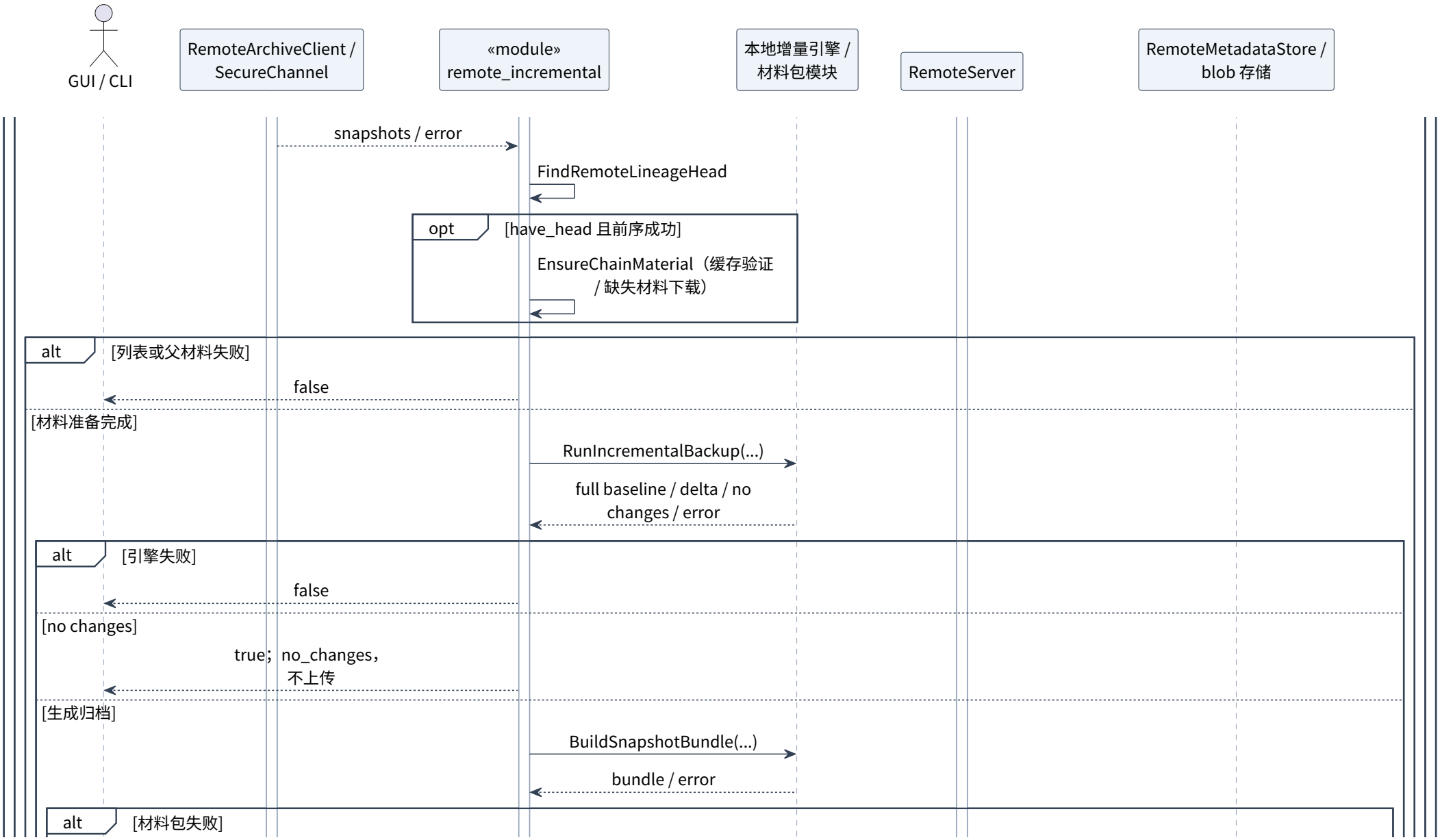
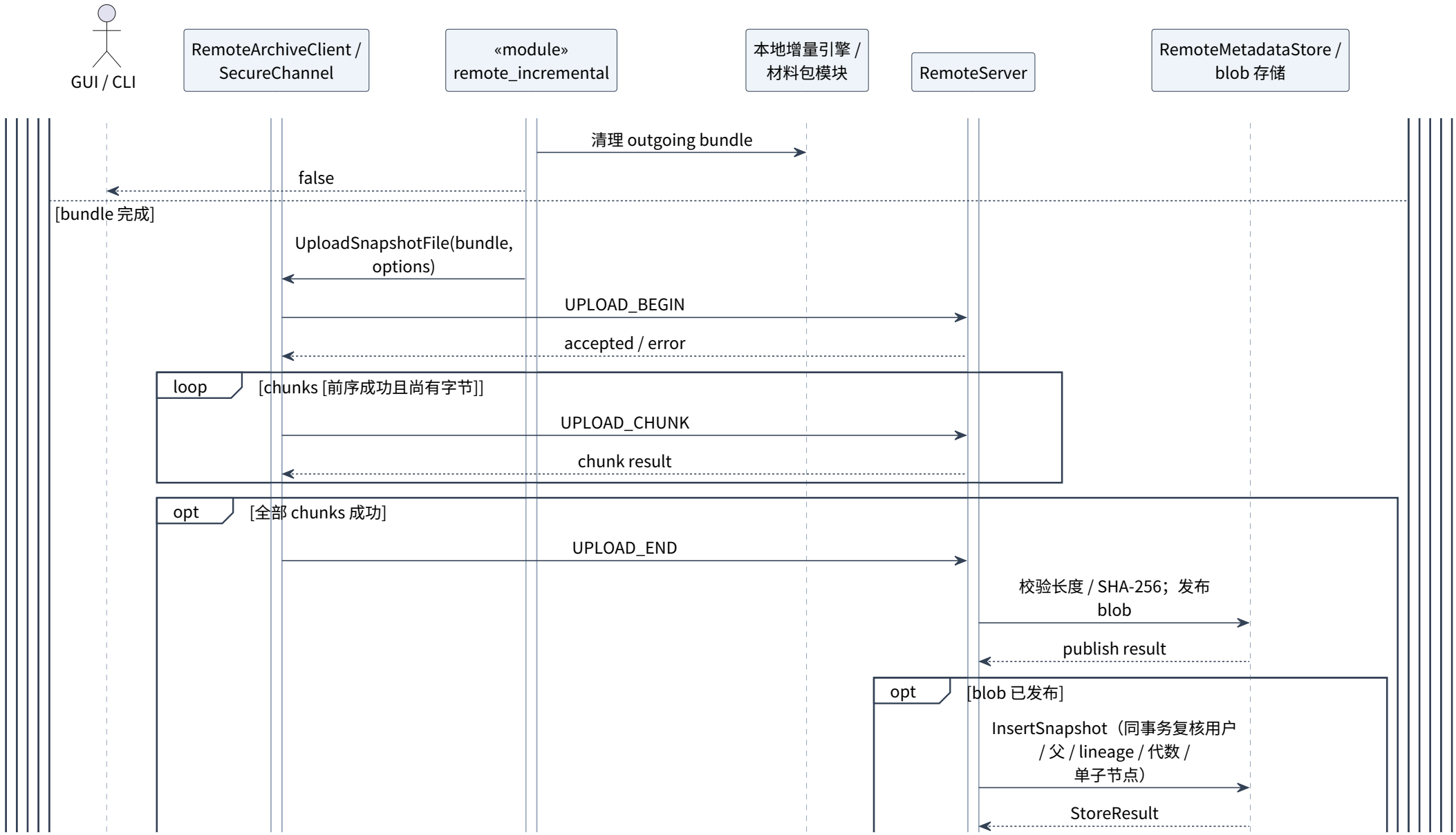
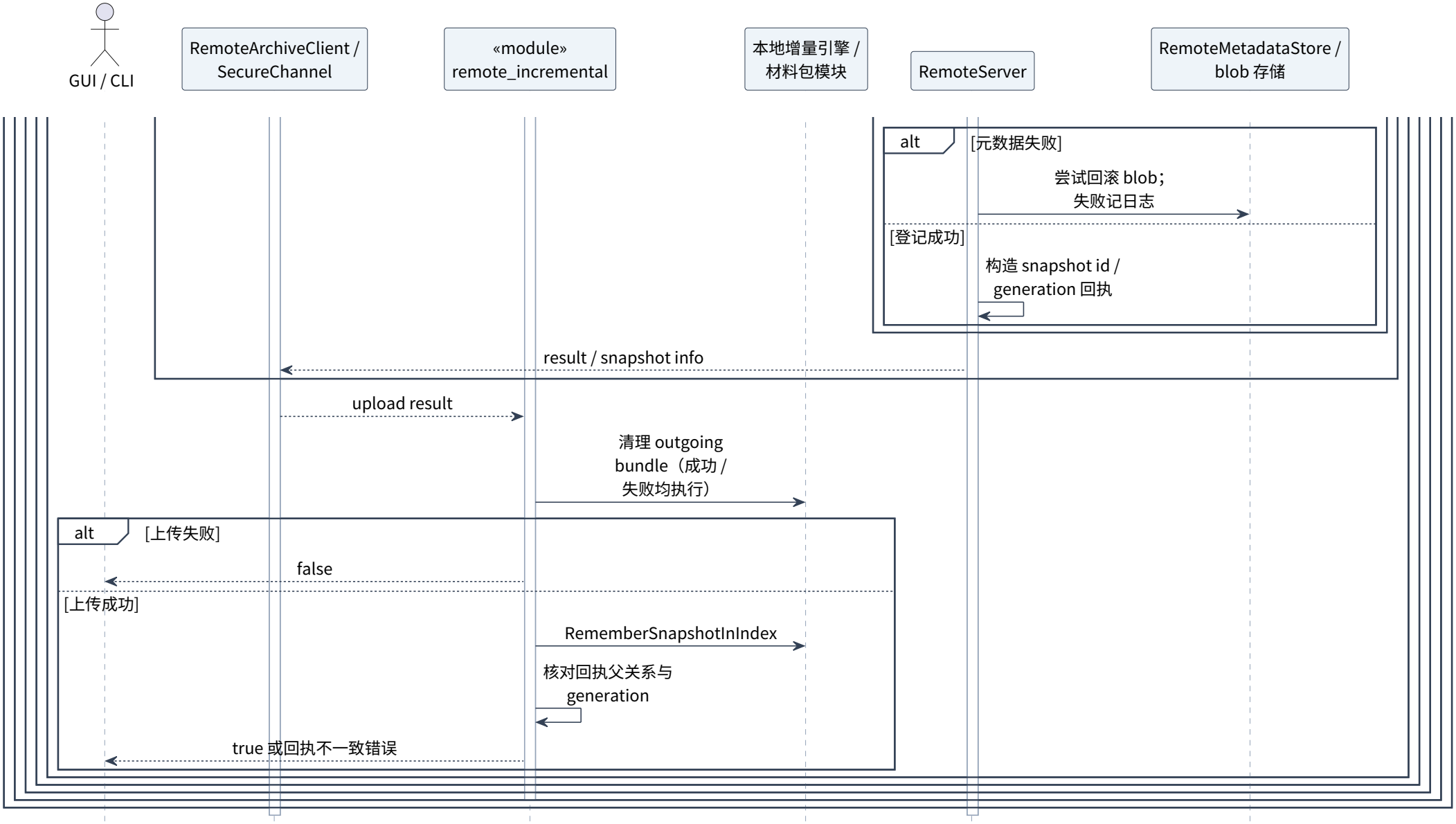

GUI / CLI
RemoteArchiveClient / SecureChannel
«module» remote_incremental
本地增量引擎 / 材料包模块
RemoteServer
RemoteMetadataStore / blob 存储
Connect(endpoint)
BPSEC1 pin / BPSEC2 certificate 握手
握手结果
alt
[身份验证失败]
false；关闭连接
[安全通道建立]
LOGIN 或 RESUME
账户请求（受 BPSEC 保护）
会话结果
result
alt
[账户失败]
中止业务请求
[会话有效]
RunRemoteBackup(request)
List()
LIST
当前账户 snapshots


GUI / CLI
RemoteArchiveClient / SecureChannel
«module» remote_incremental
本地增量引擎 / 材料包模块
RemoteServer
RemoteMetadataStore / blob 存储
snapshots / error
FindRemoteLineageHead
opt
[have_head 且前序成功]
EnsureChainMaterial（缓存验证 / 缺失材料下载）
alt
[列表或父材料失败]
false
[材料准备完成]
RunIncrementalBackup(...)
full baseline / delta / no changes / error
alt
[引擎失败]
false
[no changes]
true；no_changes，不上传
[生成归档]
BuildSnapshotBundle(...)
bundle / error
alt
[材料包失败]


GUI / CLI
RemoteArchiveClient / SecureChannel
«module» remote_incremental
本地增量引擎 / 材料包模块
RemoteServer
RemoteMetadataStore / blob 存储
清理 outgoing bundle
false
[bundle 完成]
UploadSnapshotFile(bundle, options)
UPLOAD_BEGIN
accepted / error
loop
[chunks [前序成功且尚有字节]]
UPLOAD_CHUNK
chunk result
opt
[全部 chunks 成功]
UPLOAD_END
校验长度 / SHA-256；发布 blob
publish result
opt
[blob 已发布]
InsertSnapshot（同事务复核用户 / 父 / lineage / 代数 / 单子节点）
StoreResult


GUI / CLI
RemoteArchiveClient / SecureChannel
«module» remote_incremental
本地增量引擎 / 材料包模块
RemoteServer
RemoteMetadataStore / blob 存储
alt
[元数据失败]
尝试回滚 blob；失败记日志
[登记成功]
构造 snapshot id / generation 回执
result / snapshot info
upload result
清理 outgoing bundle（成功 / 失败均执行）
alt
[上传失败]
false
[上传成功]
RememberSnapshotInIndex
核对回执父关系与 generation
true 或回执不一致错误